Kyoungmin Roh

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Wiryesunhwan-ro 100, Seongnam-si
Republic of Korea

Research Interests

I am interested in the security of complex systems that must operate reliably under evolving threats and broken assumptions. My current research focuses on CPS and system security, with a particular emphasis on learning-based intrusion detection under concept drift and the secure deployment of post-quantum cryptography in CPS and embedded systems. My research goal is to establish principled foundations for designing secure systems that remain trustworthy over time, across various environments, and during the post-quantum transition.

Work Experience

IU SEC Lab, Indiana University Bloomington *Oct 2025-Present*
Research Intern Bloomington, U.S.
Advisor: Prof. Hyungsub Kim

- Analyzing security risks and failure modes in open-source Cyber-Physical System (CPS) software
- Investigating vulnerability patterns and attack surfaces in long-lived CPS and embedded systems
- Systematically surveying CPS-related vulnerabilities using OSV, GitHub Issues, and security forums
- Developing CPS-specific vulnerability characterization to reason about real-world system security

CSOS Lab, Dankook University *Mar 2025-Present*
Research Intern Yongin, South Korea
Advisor: Prof. Seong-je Cho

- Researching learning-based intrusion detection systems (IDS) under severe concept drift without retraining
- Studying graph-based system abstractions for IDS using API co-occurrence graphs and community detection via Social Network Analysis (SNA).
- Designing adaptive decision strategies to maintain IDS reliability under evolving attack distributions
- Investigating Trusted Execution Environment (TEE)-based secure and lightweight deployment of post-quantum cryptography (PQC) with secret isolation on ARM TrustZone

Military Service, US 8th Army, 35th BDE, 2-1 ADA, E Co. *Aug 2023-Feb 2025*
KATUSA (Korean Augmentation to the U.S. Army) Waegwan, South Korea
Leadership: 1LT. John Bieganek

- Appointed as an Environmental Officer and obtained the Environmental Officer Certificate, entrusted with inspection and compliance responsibilities despite junior rank
- Conducted environmental and operational safety inspections in accordance with military compliance standards
- Earned trust from both U.S. and ROK Army leadership through reliability, procedural accuracy, and accountability
- Awarded the U.S. Army Commendation Medal (ARCOM) by a U.S. Brigadier General and the Best KATUSA Award by an ROK Army Colonel

Education

Dankook University *Mar 2021-Present*
Bachelor's in Cybersecurity Candidate Yongin, South Korea

- Undergraduate (Junior) researcher affiliated with CSOS Lab
- Relevant Coursework: AI Security, CPS Security, Embedded Security, Cryptography
- GPA: 2.7/4.0

Publications

SCI Journals

[1] **Kyoungmin Roh**, Seungmin Lee, Seong-je Cho, Youngsup Hwang, and Dongjae Kim, "SCAN: Structural Clustering with Adaptive Thresholds for Intelligent and Robust Android Malware Detection under Concept Drift," in *Computer Modeling in Engineering & Sciences (CMES)*, 2025. [SCI Q1] (Round 1 passed, Under Revision)

Domestic Journals

[2] **Kyoungmin Roh**, and Seong-je Cho, "Concept Drift-Resilient Android Malware Detection via API Co-occurrence Graphs and Louvain Communities," in *KIISE Transactions on Computing Practices (KTCP),* 2025. [Invited Paper] (Accepted)

Conferences

International Conferences

[3] **Kyoungmin Roh**, Seungmin Lee, Seong-je Cho, and Youngsup Hwang, "ALARM: Android Malware Detection with Leiden API Communities and Robust Mixture of Experts," in *The 41st ACM/SIGAPP Symposium on Applied Computing (SAC 2026),* Thessaloniki, Greece, Mar. 2026. (Accepted)

Domestic Conferences

[4] **Kyoungmin Roh**, Suhyeon Park, and Seong-je Cho, "Drift-Aware Security Module based on Louvain Communities for Retraining-Free Android Malware Detection," in *Korea Software Congress (KSC 2025)*, Yeosu, Republic of Korea, Dec. 2025 [Best Paper Award]

[5] **Kyoungmin Roh**, Seungmin Lee, Yudam Kim, Seokhyun Ahn, and Seong-je Cho, "Android Malware Detection using Co-occurrence Graphs of APIs and Louvain Method for Community Detection," in *WDSC 2025: Workshop on Dependable and Secure Computing*, Jeju, Republic of Korea, Aug. 2025 [Best Paper Award]

[6] Seungmin Lee, **Kyoungmin Roh**, Jiheon Jung, Suhyeon Park, and Seong-je Cho, "Classifying File Fragment Types for IVI System Forensics," in *WDSC 2025: Workshop on Dependable and Secure Computing*, Jeju, Republic of Korea, Aug. 2025

Research Paper Contest

[7] **Kyoungmin Roh**, Nahee Kwon, Bogyeom Kim, and Subin Jeon, "A Lightweight Secret-Isolated Post-Quantum Cryptographic Architecture for ARM TrustZone," in *Cyber Security Contest, Department of Cybersecurity, Dankook Universtiy*, Dec. 2025. [1st Place Award]

Thesis

[8] **Kyoungmin Roh**, Junhyeong Kim, and Minseok Seong, "Qrust: AI-powered QR Phishing Detection and Secure QR Generation," in *Dankook Capstone Festival, Department of Cybersecurity, Dankook University*, Dec. 2025 [Bachelor's Degree Thesis, Audience Choice Award]

Patents

[1] Title: A Malware Detection Method Combining Clustering and Supervised Learning Models
Contributors: **Kyoungmin Roh**, Seungmin Lee, Seong-je Cho, and Yoonho Choi
Korean Patent Application No. 10-2025-0098855 (Filed)

Awards & Honors

Specialized Project Scholarship *Dec 2025*
Dankook University, Department of Cybersecurity

Academic Research Scholarship *Dec 2025*
Dankook University, National Center of Excellence in Software

Cyber Security Contest 1st Place Award *Dec 2025*
Dankook University, Department of Cybersecurity

KSC 2025 Best Paper Award *Dec 2025*
Korean Institute of Information Scientists and Engineers (KIISE)

Capstone Festival Audience Choice Award *Dec 2025*
Dankook University, National Center of Excellence in Software

Dankook Startup Hackathon Encouragement Award *Sep 2025*
Dankook University, National Center of Excellence in Software

WDSC 2025 Best Paper Award *Aug 2025*
Korean Institute of Information Scientists and Engineers (KIISE)

Certificate of Appreciation *Feb 2025*
United States Army Garrison Daegu

Republic of Korea Army Soldier Award *Feb 2025*
Republic of Korea Army

The Army Commendation Medal *Dec 2024*
United States Department of the Army

Best KATUSA Award *Oct 2024*
8th United States Army Republic of Korea Army Support Group

Turing Cipher Idea Contest 1st Place Award *Sep 2023*

Dankook University, Department of Cybersecurity

Specialized Project Scholarship *Jun 2022*
Dankook University, Department of Cybersecurity

Admission Scholarship *Mar 2021*
Dankook University Admissions Office

Teaching Experiences

Basic Coding Teaching Volunteer for Middle School Students *Aug 2022*
Dankook University, Aegis Cybersecurity/Development Club

Committee Service

Dankook University, Aegis Cybersecurity/Development Club *Mar 2021-Present*
Member
- Participated in the Cybersecurity Team and CTF Team
- Contributed to collaborative security projects and CTF problem-solving

RubiyaLab CTF Team *Jul 2025-Nov 2025*
CTF Player
- Member of a top-ranked international CTF team, engaging in practical security problem-solving
- Contributed to challenges in cryptography, reverse engineering, and web security

OSINT Team *Jul 2025-Present*
Contributor
- Authored and published cybersecurity-related articles for a global OSINT education platform website

Personal Data

Languages
- Korean: Native
- English: Professional working proficiency
- Japanese: Basic communication proficiency
- Chinese: Basic Reading proficiency; limited speaking and listening

References

[1] **Prof. Seong-je Cho**
Professor, Department of Software Science, Dankook University
College of Software Convergence
Director, AI Cybersecurity Research Center
Email: sjcho@dankook.ac.kr
Phone: +82-31-8005-3229
Relationship: Academic Advisor and Research Supervisor

[2] **Prof. Hyungsub Kim**
Assistant Professor, Department of Computer Science, Indiana University Bloomington
Luddy School of Informatics, Computing, and Engineering
Email: hk145@iu.edu
Relationship: Academic Advisor and Internship Supervisor

[3] **Prof. Jiwon Seo**
Assistant Professor, Department of Cybersecurity, Dankook University
College of Software Convergence
Email: jwseo@dankook.ac.kr
Relationship: Academic Advisor and Research Supervisor